BY ORDER OF THE COMMANDER
AIR COMBAT COMMAND

AIR COMBAT COMMAND MANUAL
17-2CSCS, VOLUME 3

14 MAY 2021

Cyberspace

CYBERSPACE SECURITY AND CONTROL SYSTEM (CSCS) - OPERATIONS AND PROCEDURES

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OPR: USAF/ACC/A6OO

Certified by: USAF/ACC/A3T
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Pages: 22

This publication implements Air Combat Command Instruction (ACCI) 17-202, Volume 3, *Cyberspace Operations and Procedures.* It applies to all Cyberspace Security and Control System (CSCS) units. It applies to all ACC-assigned or attached CSCS units. This publication applies to all military and civilian Regular Air Force (AF) personnel, members of AF Reserve Command (AFRC), Air National (ANG) assigned or attached to ACC, and also to contractor support personnel in accordance with (IAW) appropriate provisions contained in memoranda support agreements and applicable AF contracts. This publication requires the collection and or maintenance of information protected by the Privacy Act of 1974, as amended, authorized by Air Force Instruction (AFI) 36-2608, *Military Personnel Records System* and System of Records Notice F011 AF AFMC B, *Patriot Excalibur (PEX) System Records*, applies and is available at **https://dpcld.defense.gov/Privacy/SORNsIndex/DOD-Component-Notices/Air-Force-Article-List/**. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW Air Force Records Information Management System Records Disposition Schedule. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance

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Chapter 1

ROLES AND RESPONSIBILITIES

1.1. Overview/Background. This volume, in conjunction with other governing directives, prescribes procedures for operating the CSCS weapon system under most circumstances. It is not a substitute for sound judgment or common sense. Procedures not specifically addressed may be accomplished if they enhance safe and effective mission accomplishment.

1.2. General Guidance.

1.2.1. Refer to ACCI 17-202, Volume 3, *Cybercrew Operations and Procedures*, for all Major Command (MAJCOM) and Numbered Air Force (NAF) roles, responsibilities and programs. For the purposes of this manual, "MAJCOM" includes direct reporting units, Air Force Reserve Command (AFRC) and the National Guard Bureau (NGB).

1.2.2. Refer to ACCI 17-202, Vol 3, for all unit-level roles, responsibilities and programs. For purposes of this manual, "unit" includes levels of organization under Higher Headquarters (HHQ) required to establish a cybercrew function.

1.3. Deviations. In the case of an urgent requirement or emergency, the Crew Commander will take appropriate actions(s) to ensure safe operations. **(T-3).**

Chapter 2

MISSION PLANNING

2.1. Responsibilities. Individual crews, unit operations and intelligence functions jointly share responsibility for mission planning. The Crew Commander is ultimately responsible for ensuring all aspects of mission planning, to include complying with command guidance, is accomplished. Unit commanders may supplement mission-planning requirements but will ensure an appropriate level of mission planning is conducted prior to each mission. All missions and/or events will follow the Planning, Briefing, Execution, and Debriefing (PBED) Guide or mission planning guidelines. **(T-3).**

2.2. Mission Planning Guidelines.

2.2.1. Effective mission accomplishment requires thorough mission planning and preparation. Specific mission planning elements are addressed in Air Force Tactics, Techniques, and Procedures (AFTTP) 3-3, *Integrated Planning and Employment*, AFTTP 3-1, *General Planning*, Joint Cyber Tactics Manual (JCTM), 688 Cyberspace Wing Tactical Mission PBED Guide, and any local crew aids. While these manuals are authoritative, they are not directive; however, they are useful in ensuring adequate mission planning and employment.

2.2.2. Commanders will provide adequate time and facilities for mission planning. Crews will accomplish sufficient planning to ensure successful mission accomplishment. **(T-3).** Units will maintain facilities where all information and materials required for mission planning are available. **(T-3).**

2.2.3. Commanders will ensure other activities, (e.g., recurring academic training, training device periods, additional duties), do not interfere with time allotted for mission planning and crew mission briefing. **(T-3).**

2.2.4. ACCI 17-202, Vol 3, mandates that, “Units will accomplish sufficient planning to ensure successful mission accomplishment for all phases of operations.” All units operating the CSCS weapon system will ensure compliance with the ACCI’s minimum mission planning requirements. **(T-3).**

2.2.5. As a result of this planning, units may create an aid for execution. These should, at a minimum, include the Tasking Order and line number, Crew accountability, the Area of Operation/Terrain, Communication plan, Vulnerability/operating window, and Deconfliction plan.

2.3. Briefings.

2.3.1. The Crew Commander is responsible for presenting a logical briefing to promote a safe and effective mission(s). All crewmembers will attend the mission brief unless previously coordinated with the Squadron Director of Operations (SQ/DO) or designated representative. **(T-3).**

2.3.2. The Crew Commander will plan adequate time to discuss required briefing items commensurate with the complexity of the mission and operator capabilities. Any item published in MAJCOM/NAF/wing/group/squadron standards or AFIs that is understood by all participants may be briefed as "standard". **(T-3).**

2.3.3. Briefings should conclude 15 minutes prior to scheduled sortie/crew shift.

2.3.4. Briefing Guides. Briefing guides will be used by the lead briefer. An example briefing guide can be referenced in **Attachment 2**. Units may augment these guides as necessary. Items may be briefed in any sequence, provided all minimum requirements listed in this ACCMAN and other local directives and guidance are addressed. **(T-3).**

2.3.5. All briefings will include:

2.3.5.1. Risk Management (Mission and Personnel Risk Assessments) **(T-3).**

2.3.5.2. Mission/Event and crew Go/No-Go status **(T-3).**

2.3.5.3. Mission/Event priorities and objectives **(T-3).**

2.3.5.4. Crew accountability **(T-3).**

2.3.5.5. Abort Criteria, Alternate Mission/event, and Contingency Plans **(T-3).**

2.3.5.6. Deconfliction, Environment, Terrain **(T-3).**

2.3.5.7. Taskings (e.g., Special Instructions (SPINs), Cyber Tasking Orders, Rules of Engagement (ROE)) **(T-3).**

2.3.5.8. Roles, and responsibilities of each crewmember **(T-3).**

2.3.5.9. Weapon system status, and facility status **(T-3).**

2.3.5.10. Threat Intelligence

2.3.5.10.1. Unit-level intelligence will provide relevant threat intelligence identifying adversary intent, tools utilized, and tactics performed to support mission planning practices and mission execution. In the absence of organic unit-level intelligence, wing intelligence will provide intel briefing support to lead briefer or mission commander by request. **(T-3).**

2.3.6. Anyone not attending the mission brief will receive, at a minimum, an overview of the mission/event objectives, their roles and responsibilities and emergency procedures (EP) prior to becoming actively involved in crew operations. **(T-3).**

2.3.7. Positional Changeover Brief. For operational needs, crews will be required to brief oncoming crewmembers. These will be delivered IAW local briefing guides. **(T-3).**

2.3.8. Alternate Missions/events. Alternate missions will be briefed in case the originally planned and briefed mission is cancelled/aborted. **(T-3).**

2.3.8.1. Mission/event components may be modified and briefed after start of execution as long as mission safety is not compromised. Crew Commanders will ensure changes are acknowledged by all crewmembers. **(T-3).**

Chapter 3

NORMAL OPERATING PROCEDURES

3.1. Pre-Mission Arrival Times. The SQ/DO may adjust crew report time to meet mission requirements. Crew report times will allow sufficient time to accomplish all pre-mission activities. **(T-3).**

3.2. Pre-Mission Duties. Prior to mission execution, crewmembers should only be scheduled for duties related to the scheduled mission/event. **(T-3).**

3.2.1. Example: Crewmember is scheduled for a sortie/crew shift from 0830 – 1430; crewmember duty day is scheduled from 0730 – 1630. Prior to mission execution, crewmember performs pre-mission duties (e.g., planning, briefing). After the mission debrief, crew is released to perform other/additional duties. **(T-3).**

3.3. Crew Information File (CIF). Unit Standardization and Evaluation (Stan/Eval) will ensure accurate CIFs are available for crewmembers. Crewmembers will review CIFs before all missions/events. **(T-3)**.

3.3.1. Crewmember review will be documented in the MAJCOM approved automated software. If automated functionality is not available, to accomplish the CIF review and documentation, units will use unit Stan/Evals backup CIF procedure. **(T-3).**

3.3.2. Electronic signatures or sign-off will be used on CIFs. **(T-3).**

3.4. Go/No Go. The Squadron Commander (SQ/CC) will implement the Go/No-Go program to ensure individual crewmembers are current, qualified and/or adequately supervised to perform operations. Unit Stan/Eval offices are responsible for management of the overall Go/No-Go program. Crewmembers will not operate on the weapon system until the Go/No-Go is accomplished and verified. **(T-3).**

3.4.1. The Crew Commander is responsible for accomplishing the daily Go/No-Go verification for all crewmembers performing mission duties for that sortie/crew shift. **(T-3).**

3.4.2. Designated individuals will verify, document, and sign off on the Go/No-Go status prior to releasing crewmembers for any mission/event execution. Go/No-Go accomplishment is an essential item in the mission pre-brief. Record the Go/No-Go accomplishment and verification in the MAJCOM approved automated system (e.g., Patriot Excalibur (PEX)). **(T-3).**

3.4.3. If automated functionality is not available, to accomplish the Go/No-Go verification, units will implement unit Stan/Evals developed backup procedures to permit Go/No-Go verification when the relevant information system is unavailable. **(T-3).**

3.4.4. The unit Go/No-Go process will verify the following minimum items for all crew members scheduled to perform crew duties:

3.4.4.1. Qualification/certification IAW ACCMAN 17-2CSCS Volume 1, *Cyberspace Security and Control System (CSCS) Cybercrew Training*, and ACCMAN 17-2CSCS Volume 2, *Cyberspace Security and Control System (CSCS) Standardization and Evaluations*, for the crew position, mission/event, and duties they are scheduled to perform. **Note**: Crewmembers not qualified/certified and in training status will require instructor or evaluator supervision to conduct crew duties. **(T-3).**

3.4.4.2. Individual Risk Assessment. Units will develop procedures to conduct individual assessments to assess their sleep adequacy before assigned shift, their personal stress level, their health/safety concerns, and their circadian disruption. It is an individual's responsibility to notify a Crew Commander on shift when a physical or mental condition exists, affecting themselves or other unit personnel, that may prevent mission accomplishment, cause mission degradation, or endanger personnel safety. **(T-3).**

3.4.4.3. Ensure currency, IAW Ready Cybercrew Program (RCP) Tasking Memorandum (RTM), and in conjunction with ACCMAN 17-2CSCS Vol 1 for the crew position, mission/event, and duties they are scheduled to perform. **Note**: Crewmembers not current in the lookback sorties/crew shifts, within their crew position, will require instructor supervision to conduct crew duties until regaining currency. **(T-3).**

3.4.4.4. Crew Information File (CIF). Document review by each crewmember of Volume 1, Part B, CIF items. **Note**: A review of all volumes will be accomplished prior to an individual receiving their first operational training. Assigned or attached crewmembers on extensive absence from conducting missions (90 days or more) will accomplish a complete review of all volumes prior to operations. **(T-3).**

3.5. Crew Changeover. Crewmembers from the off-going and on-coming sorties/crew shifts will participate in a crew changeover briefing. Direct crew changeover only applies if crews are performing 24/7 operations. If 24/7 operations are not occurring, crew changeover briefings will be incorporated into the next mission brief. At a minimum, the changeover will include all items previously identified for the mission brief. **(T-3).**

3.6. Unit-Developed Checklist/Local Crew Aids.

3.6.1. Locally developed checklists and crew aids may be used, and will not supersede technical guidance or Technical Orders. At a minimum, include the following:

3.6.1.1. EP checklists and communication-out information. **(T-3).**

3.6.1.2. Other information as deemed necessary by the units (e.g., local mission planning guides, briefing/debriefing guides, reference data sheets, local training diagrams, local area maps of sufficient detail to provide situational awareness on area boundaries). **(T-3).**

3.6.2. Unit Stan/Eval, Operations Training, and Weapons and Tactics offices will review prior to approval. **(T-3).**

3.6.3. Unit Stan/Eval office will maintain the list of current and authorized checklists, crew aids and other information as necessary in the CIF library. **(T-3).**

3.7. Forms and Logs. The Master Station Log (MSL) is the unit's official record of events that occurred during operations or training. The purpose is to maintain an accurate and detailed record of all significant events pertaining to operations occurring during each sortie. Each crew position operator is responsible for documenting significant events/crew actions required for the MSL. The Crew Commander is responsible for content, accuracy and timeliness of all inputs to mission-related information management portals/collaborative information sharing environments IAW with applicable directives, tasking and policy. **(T-3).**

3.8. Required Equipment/Publications. Crewmembers will have all equipment and publications required for mission/event execution. These may be maintained and carried

electronically provided operable viewing and printing capability exists throughout mission/event execution. OGV will maintain the list of required publication items. **(T-3).**

3.9. Operations Check (Ops Check).

3.9.1. Crew members will perform Ops Checks at beginning of sortie/crew shift, and as required during sortie/crew shift period based on mission/event triggers and requirements. **(T-3).**

3.9.2. Crews will, at a minimum, check the following items during Ops Checks: verification of access and operational mission equipment status. **(T-3).**

3.10. Vulnerability Window. Crews are bound by the vulnerability window. Deviations from the assigned vulnerability window will be coordinated through the Crew Commander and approved by the tasking authority. **(T-3).**

3.11. Dynamic Retasking/Targeting. Ad hoc and/or emerging target tasking can occur. During tasked missions, an operator may identify and report something that may require ad hoc or dynamic re-tasking/targeting. Dynamic re-tasking allows modification of the mission to support changing mission objectives. This includes everything from re-tasking the operator within the mission requirements to an entirely new mission that is added to the current tasking order. **(T-3).**

3.12. Crew Coordination. Is applied during mission events, scenario or tasks which require interaction between all applicable crew members to ensure situational awareness. During crew coordination, the crew has the ability to monitor a particular status and associate the impact of the action. **(T- 3).**

3.12.1. Two-person Integrity (TPI), is mandatory for CSCS operators conducting medium/high risk actions as determined by risk management assessment during mission planning. Policies and procedures must be followed to ensure changes and actions are properly reviewed and approved prior to execution to avoid adverse effects. SQ/DO and Crew Commander will enforce TPI standards for crew operations. **(T-3)**.

3.13. Communications.

3.13.1. Mission execution requires at least one method of communication for all operations. **(T-3).**

3.13.2. The Crew Commander is responsible for developing and briefing the communication plan. Communications planning is performed to determine who to contact and the medium available to use. An example Communications Planning Guide can be found in **Attachment 4**. **(T-3).**

3.14. Mission Report (MISREP). Tasking authorities, future missions and debriefs rely on timely, accurate MISREPs. All MISREPs will be completed prior to debrief. **(T-3).**

3.14.1. The Crew Commander is responsible for providing timely, accurate and correctly formatted reports to the tasking authority. **(T-3).**

3.14.2. A MISREP will be accomplished once the cybercrew completes a mission or particular phase of the mission IAW guidance/tasking. **(T-3).**

3.14.3. The Crew Commander will review the MISREP for accuracy and completeness, and submit to the tasking authority. **(T-3).**

3.14.4. Each crewmember is responsible for providing the appropriate data regarding their mission area for the MISREP. **(T-3).**

3.14.5. Local procedures/templates may be developed to ensure standardization of reporting.

3.15. Debriefing.

3.15.1. The Crew Commander is responsible for leading the crew debrief. **(T-3)**.

3.15.2. Debriefs will address all aspects of the mission (planning, briefing, and execution) and ensure all participants receive feedback through the development of Lessons Learned (LL). LL should be applied as soon as operationally feasible (e.g., Weapons and Tactics, Stan/Eval and Operations Training involvement which may result in process changes and/or updated checklists). SQ/DO will determine a tracking process to ensure LL are implemented. **(T-3).**

3.15.3. Debrief will address crewmember responsibilities, de-conflictions, tactical employment priorities, and task management. **(T-3).**

3.15.4. The debriefing should occur at multiple levels (e.g., crew debrief, positional debrief) depending on the mission/event type (e.g., Line of Effort, degradation loss). The debriefing guide is in **Attachment 3**.

Chapter 4

CREW DUTIES, RESPONSIBILITIES AND PROCEDURES

4.1. CSCS Crew Requirement. The CSCS Crew will consist of expertise from each component of the weapon system (see AFTTP 3-1, CSCS Crew Construct). This enables the CSCS Crew to employ all aspects of the CSCS Weapon System and support all CSCS mission types. In some cases, there may be personnel with expertise that may support multiple mission/events. **(T-3).**

4.2. Crew Stations. Crewmembers will be on the operations floor during the critical points/execution as determined by the Crew Commander. Crewmembers will notify the Crew Commander prior to departing their assigned primary duty location. **(T-3).**

4.3. Crew Duties. Crews are responsible to the Crew Commander for the safe, effective use of the weapon system and mission accomplishment. A crew brief will be accomplished before each mission execution to ensure an understanding of all aspects of the mission. **(T-3).**

4.4. Crew Positions.

4.4.1. Crew Commander (CSCS-CC). Serves as the command authority for CSCS crew operations and provides command oversight for operations crew personnel. The CSCS-CC is also responsible for the management of crewmembers' execution of assigned missions as well as enforcing policies and procedures to ensure successful mission accomplishment. Crew Commander responsibilities and/or authority include;

4.4.1.1. Managing crew resources and ensuring the safe and effective conduct of operations and mission accomplishment. **(T-3).**

4.4.1.2. Welfare of the crew. **(T-3).**

4.4.1.3. Ensuring that any portion of the operation affecting the accomplishment of the mission is coordinated with the tasking authority. **(T-3).**

4.4.1.4. Ensuring risk management decisions are utilized through all phases of the mission. **(T-3).**

4.4.2. Operations Controller (CSCS-OC). Responsible for management of operations, execution of assigned missions and reporting their status to Crew Commander.

4.4.3. Operations Crew Personnel. Operate one or more components of the CSCS Weapon System executing operations as tasked by the Crew Commander to support friendly forces and accomplish mission objectives in assigned terrains as outlined in assigned mission plans. CSCS Operator positions include:

4.4.3.1. Client End Point Protection Operator (CSCS-CPO). Remediates vulnerabilities and weaknesses identified in cyberspace terrain and associated software suites utilized by Air Force Information Systems and net-centric capabilities.

4.4.3.2. Directory Services Operator (CSCS-DSO). Provides authentication and accessibility to clients in the Air Force Network domain.

4.4.3.3. Storage and Virtualization Operator (CSCS-SVO). Performs backup, recovery and archiving via storage area networks (SAN).

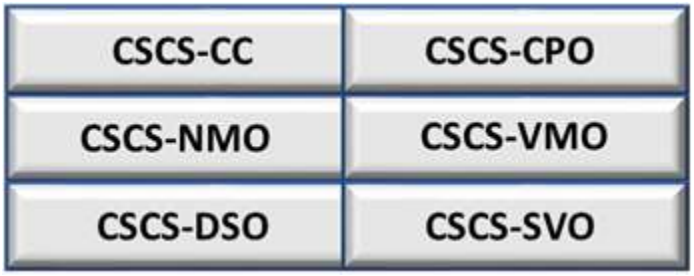
4.4.3.4. Vulnerability Management Operator (CSCS-VMO). Identifies and remediates vulnerabilities within cyberspace terrain and associated software suites utilized by AF information systems and net-centric capabilities. In addition, identifies and assesses the weaknesses in cyberspace terrain that adversaries may exploit to gain/maintain Air Force Network (AFNet) access. This new crew position is a combination of CSCS-VAO and CSCS-VRO.

4.4.3.5. Network Management Operator (CSCS-NMO). Allows, denies, redirects, and logs network traffic in, thru, and from base firewalls, proxies, routers, and switches. In addition, the CSCS-NMO evaluates, detects, prevents, and implements counter-measures to protect network hosts, data, voice, and key mission systems from unauthorized network activity. The following items are the minimum standards for individuals assigned to the Network Management crew positions. This new crew position is a combination of CSCS-BPO and CSCS-IFO.

4.5. Crew Construct. A standard full complement CSCS Crew consists of six crew positions. The crew construct may vary based on the type of mission; SQ/DO may tailor crew manning to meet operational requirements. CSCS operating units that cannot field a full complement crew are required to identify their crew construct to OGV. See **Figure 4.1** for full complement crew construct positional requirements. **(T-3). Note**: CSCS is currently going through a phased crew position consolidation approach, which will change the standard full complement.

4.5.1. The 690 NSS is the only squadron operating the CSCS with a crew compliment consisting of just two crew positions: Crew Commander and Operations Controller.

Figure 4.1. Standard Full Complement CSCS Crew Construct.



4.5.2. The 690th Network Support Squadron (NSS) is the only squadron operating the CSCS with a crew compliment consisting of just two crew positions: Crew Commander and Operations Controller see **Figure 4.2**

Figure 4.2. 690 NSS Standard Full Complement CSCS Crew Construct.



4.6. Maximum Load out per CSCS Crew per Sortie/Crew shift for missions. (T-3).

4.6.1. Limitations based on capacity and mission types will be identified by HHQ's mission planning based on the number of CSCS crews available for tasking. **(T-3).**

4.6.1.1. Due to the varying scope of missions, SQ/DOs and CSCS Crews will coordinate with HHQs to identify additional CSCS crew ops capacity during mission planning. **(T-3).**

4.6.1.2. Mission planning should identify capacity constraints, restraints, and the minimum crew construct required. **(T-3).**

4.7. New/Modified Equipment and/or Capabilities. Crewmembers not qualified and/or certified in the operation of new or modified equipment and/or weapon system capabilities will not operate that equipment or perform any duties associated with that weapon system capability unless under the supervision of a current and qualified instructor. The 690th Cyber Operations Group Stan/Eval (COG/OGV) will spearhead standardization of new/modified equipment and/or capabilities within the CSCS Stan/Eval and Operations Training programs (e.g., Type 1 Training). **(T-3).**

4.8. Crew Rest/Duty Period/Sortie Crew Shift Duration. Crew rest, crew duty period and crew augmentation will be IAW ACCI 17-202 Vol 3 and all applicable guidance with the following additional guidance:

4.8.1. Crew Rest. Commanders will ensure crews are afforded a minimum 10-hour non-duty period before the duty period begins to ensure the crewmember is provided adequate time to rest before performing a mission or mission-related duties. Crew rest is free time, and includes time for meals, transportation and 8 hours of uninterrupted sleep. Rest is defined as a condition that allows an individual the opportunity to sleep; however, each crewmember is individually responsible for ensuring they obtain sufficient rest during the provided crew rest periods. **(T-3).**

4.8.2. Duty Period Length. Maximum duty period lengths are addressed in ACCI 17-202 Vol 3, Section 4.1.

4.8.2.1. Continuous operations means three or more consecutive sorties of at least 12 hours duration separated by minimum crew rest. **(T-3).**

4.8.2.2. The crew rest exception shall only be used for contingency/surge operations and not for scheduling conveniences. **(T-3).**

4.8.2.3. The Crew Commander will ensure any reduction in the crew rest period that was approved by the SQ/CC or COG/CC is annotated in the MISREP. **(T-3).**

4.8.3. Duty Period. The crew duty day will not exceed 12 hours, which includes planning, briefing and debriefing. The SQ/DO must approve any requests for a crewmember to perform or attend non-mission related duties or events during a phase of the PBED process. **(T-3).**

4.8.4. Sortie/Crew Shift duration. For planning purposes, the average sortie duration (ASD) or crew shift duration is 8 hours.

4.9. Crew Force Management. The goal to crew force management is to maximize crew force experience while sustaining staff manning at an adequate level for unit programs.

4.9.1. Crew Scheduling (DOS). Units will establish a crew scheduler to manage mission crew scheduling. The crew scheduler is the focal point to crew force management. The schedule will be accomplished IAW crew rest limitations outlined in **paragraph 4.9**. Units will publish, post, and monitor schedules for the crew force and initiate changes to the schedules based on

proper tracking of qualifications, certifications, restrictions and other factors as required to meet mission objectives. **(T-3).**

4.9.1.1. Crew Commanders will ensure a crewmember on leave or temporary duty (TDY) is notified if an event is altered in any way or added to the crewmember's schedule within the first 72 hours of their scheduled return. **(T-3).**

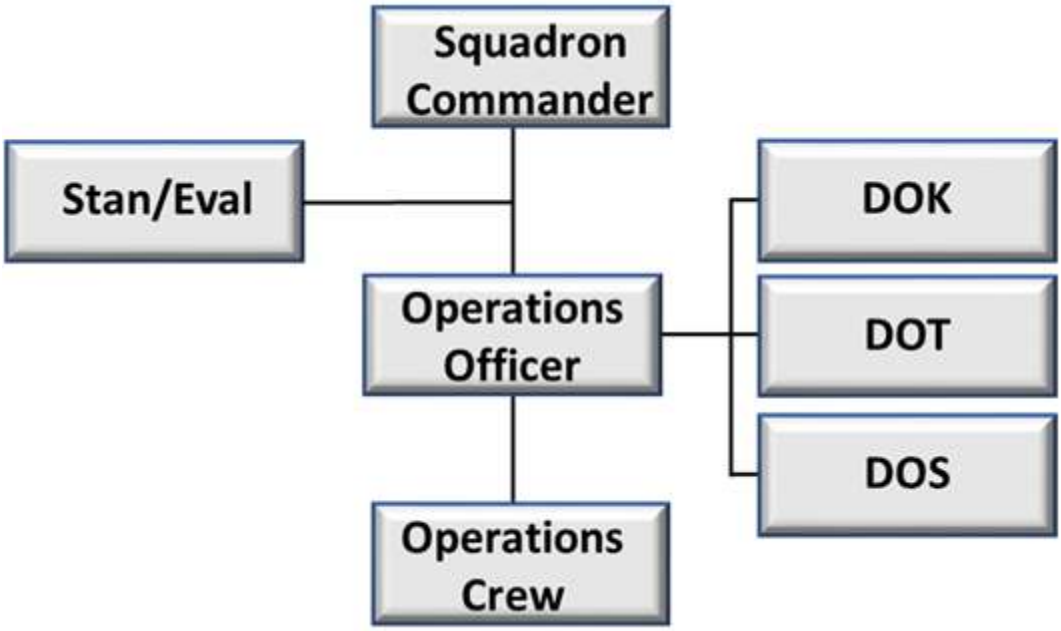
4.9.1.2. Notifications will be made immediately after the change is official, no later than 12 hours prior to the scheduled event start time. Units will ensure that oncoming crewmembers will be capable of meeting crew risk management requirements. **(T-3).**

4.9.1.3. Under unusual circumstances (e.g., sudden illness, increased operational activity), the SQ/CC or operations officer may place a crew member on operational duty at any time, provided they have had the opportunity for crew rest.

4.9.1.4. Crew Scheduling will work with Stan/Eval and Training to incorporate their schedules into a comprehensive crew schedule. Scheduling Stan/Eval and Operations Training occurrences, on the operations floor, will not disrupt operations. The scheduler will deconflict and adhere to safe operations when managing the crew schedule. **(T-3).**

4.9.2. CSCS Operations Squadron Construct. Provides CSCS weapon system crews, Stan/Eval, Operations Training, Weapons and Tactics, crew scheduling, and administration. See **Figure 4.3** for the CSCS Operations Squadron Construct. **(T-2).**

Figure 4.3. CSCS Operations Squadron Construct.



MARK H. SLOCUM, Maj Gen, USAF
Director of Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

ACCI 17-202, Volume 3, *Cyberspace Operations and Procedures*, 8 October 2019

ACCMAN 17-2CSCS, Volume 1, *Cyberspace Security and Control System (CSCS) - Cybercrew Training*, 14 May 2021

ACCMAN 17-2CSCS, Volume 2, *Cyberspace Security and Control System (CSCS) - Standardization/Evaluation Program*, 14 May 2021

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFI 36-2608, *Military Personnel Records System*, 26 October 2015

AFTTP 3.1.General Planning

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Prescribed Forms

None

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

ACC—Air Combat Command

ACCI—Air Combat Command Instruction

ACCMAN—Air Combat Command Manual

AF—Air Force

AFCYBER—Air Forces Cyber

AFNET—Air Force Network

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFRC—Air Force Reserve Command

AFTTP—Air Force Tactics, Techniques and Procedures

ANG—Air National Guard

ASD—Average Sortie Duration

BCQ—Basic Cybercrew Qualification

BMC—Basic Mission Capable

CC—Commander

CIF—Crew Information File

COG—Cyberspace Operations Group

COG/OGV—Cyber Operations Groups / Standards & Evaluations

CSCS—Cyberspace Security and Control System

CSCS—BPO—CSCS Boundary Protection Operator

CSCS—CC—CSCS Crew Commander

CSCS—CPO—CSCS Client End Point Protection Operator

CSCS—DSO—CSCS Directory Services Operator

CSCS—IFO—CSCS Infrastructure Operator

CSCS—NMO—CSCS Network Management Operator

CSCS—OC—CSCS Operations Controller

CSCS—SVO—CSCS Storage and Virtualization Operator

CSCS—VAO—CSCS Vulnerability Assessment Operator

CSCS—VMO—CSCS Vulnerability Management Operator

CSCS—VRO—CSCS Vulnerability Remediation Operator

CT—Continuation Training

DAFI—Department of the Air Force Instruction

DO—Director of Operations

DoDIN—Department of Defense Information Network

DOS—Crew Scheduling

EP—Emergency Procedures

HHQ—Higher Headquarters

IAW—In Accordance With

IP—Internet Protocol

IQT—Initial Qualification Training

JFHQ-C—Joint Forces Headquarters-Cyber

LL—Lessons Learned

MAJCOM—Major Command

MISREP—Mission Report

MQT—Mission Qualification Training

MR—Mission Ready

MSL—Master Station Log

NAF—Numbered Air Force

NGB—National Guard Bureau

NSS—Network Support Squadron

OGV—Standardization and Evaluation

PBED—Plan, Brief, Execute, Debrief

PEX—Patriot Excalibur

OPR—Office of Primary Responsibility

Ops Check—Operations Check

RCP—Ready Cybercrew Program

ROE—Rules of Engagement

RTM—RCP Tasking Memorandum

SAN—Storage Area Network

SPINs—Special Instructions

SQ—Squadron

SQ/CC—Squadron Commander

SQ/DO—Squadron Director of Operations

Stan/Eval—Standardization and Evaluation

TDY—Temporary Duty

TPI—Two-person Integrity

URL—Universal Resource Locator

USAF—United States Air Force

Terms

Average Sortie/Crew Shift Duration (ASD)—ASD is used to convert sorties to operating/execution hours and vice versa. MAJCOM/A3TB uses the unit's last programmed ASD when initially determining execution/operating hours programs for the current and future years. Units will update ASD annually to reflect the unit's best estimate of the optimum sortie duration after considering historical experiences, changes in missions, deployments, etc. The formula to calculate ASD is ASD = # of weapon system hours employed divided by number of sorties**.**

Basic Cybercrew Qualification (BCQ)—A cybercrew member who satisfactorily completed IQT. The crewmember will have BCQ status only until completion of MQT. BCQ crewmembers will not perform RCP-tasked events or sorties without instructor supervision.

Basic Mission Capable (BMC)—A cybercrew member who has satisfactorily completed IQT, MQT, and a full Stan/Eval evaluation, and maintains certification, currency and proficiency in the command or unit operational mission, is BMC. The cybercrew member must be able to attain Mission Ready (MR) status to meet operational taskings as specified in the applicable weapon system volumes. Persons filling a BMC position must be able to meet MR currency requirements within 30 days (90 days for Air Reserve Component).

Certification—Designation of an individual by the certifying official as having completed required training and/or evaluation and being capable of performing a specific duty.

Continuation Training (CT)—Training, which provides crewmembers with the volume, frequency and mix of training necessary to maintain currency and proficiency in the assigned qualification level.

Crew—Also referred to as crewmembers, consist of individuals who conduct cyberspace operations or computer network exploitation and are typically assigned to a specific weapon system.

Crew Commander—The Crew Commander serves as the command authority for CSCS crew operations and provides command oversight for operations floor personnel as well as enforcing policies and procedures to ensure successful mission accomplishment.

Crew Information File (CIF)—A collection of publications and material determined by the MAJCOM and unit as necessary for day-to-day operations.

Currency—A measure of how frequently and/or recently a mission/event/task is completed. Currency requirements should ensure the average crewmember maintains a minimum level of proficiency in a given event.

Cyber (adjective)—Of or pertaining to the cyberspace environment, capabilities, plans or operations.

Cyberspace—A global domain within the information environment consisting of the interdependent network of information technology infrastructures and resident data, including the Internet, telecommunications networks, computer systems, and embedded processors and controllers.

Cyberspace Operations (CO)—The employment of cyberspace capabilities where the primary purpose is to achieve objectives in or through cyberspace.

Event—An item that occurs or is encountered that initiates a process requiring a set of tasks to be accomplished. Multiple events may be completed and logged during a sortie (be it operational sortie or a training sortie) unless specifically excluded elsewhere in this instruction**.**

Mission—Missions are operations conducted with an intended purpose. Missions are conducted by a unit and/or units with relevant capability and preponderance of capacity. The base mechanism used to achieve mission objectives are sorties. Missions may require multiple sorties from multiple units to accomplish the mission’s objectives.

Mission Ready (MR)—A cybercrew member who has satisfactorily completed IQT and MQT, Stan/Eval evaluation, and maintains qualification, currency and proficiency in the command or unit operational mission is MR.

Qualification—Qualifies a crewmember to perform the duties of a particular crew position in the specified weapon system. Requires AF Form 8/8a documentation.

Ready Cybercrew Program (RCP)—RCP is the formal continuation training (CT) program that provides the baseline for squadrons to use in developing a realistic training program to meet all Designed Operational Capabilities statement tasked requirements as well as specific NAF mission prioritization. RCP defines the minimum required mix of annual sorties/crew shifts, simulator/training environment, and training events crews will accomplish to sustain mission readiness. These programs have clearly defined objectives and minimum standards that enhance mission accomplishment and safety. RCP sorties/crew shifts and events are tracked. RCP applies to MR and BMC positions**. Sortie**—The actions an individual weapon system takes to accomplish a mission and/or mission objective(s) within a defined start and stop period**.**

Tasking—A clearly defined action or activity specifically assigned to an individual or organization that will be done as it is imposed by an appropriate authority**.**

Terrain—The cyberspace area of operations where a force package is directed to conduct a sortie. Terrain is defined as Internet Protocol (IP) address, domain, or transport space within the DoDIN or AF enclave (commonly referred to as "blue" space), or commercial, contractor-owned, mission partner-owned ("gray" space) host, server, and network devices that enable C2, communication, sensing, and access capabilities**.**

Vulnerability Window—This is a window of opportunity and direction for a tactical commander to conduct operations. A vulnerability window is bounded (start by/finish by) to give a tactical commander the authorized and suspensed timing available to plan and prosecute mission. Deviations from the assigned vulnerability window must be approved by the tasking authority**.**

Weapon System—A combination of one or more weapons with all related equipment, materials, services, personnel, and means of delivery and deployment (if applicable) required for self-sufficiency.

Attachment 2

EXAMPLE BRIEFING GUIDE

A2.1. Briefing Guides. This attachment provides items for guidance and consideration when developing unit briefing guides. Additional guidance and information can be found in the 16 AF, *AFCYBER & JFHQ-C AFCYBER Tactical Mission Planning, Briefing and Debriefing Guide*, and others. These manuals are authoritative, not directive, and should be considered when developing unit specific guides.

Table A2.1. Example Briefing Guide.

☐ Time Hack
☐ Administration
☐ Crew Accountability
☐ Classification
☐ Mission/Event Assignments
☐ Mission / Tasking
☐ Environmental
☐ Capabilities
☐ Effects
☐ Plan
☐ Abort Criteria / Contingencies
☐ Comm Plan / C2 Plan
☐ Reporting
☐ Alternate Mission
☐ Questions / Comments
Note: This layout can be used for multiple briefs; however, the focus areas and emphasis items will be different as the audiences are different.

Attachment 3

EXAMPLE DEBRIEFING GUIDE

Table A3.1. Example Debriefing Guide.

☐ Debrief Presentation Format:
☐ Classification
☐ ROE
☐ Admin / Alibis
☐ Objectives
☐ Planning
☐ Execution
☐ Big Rocks
☐ Debrief Focus Points
☐ Lessons Learned
☐ Assessments

Attachment 4

EXAMPLE COMMUNICATIONS PLANNING GUIDE

Table A4.1. Example Communications Planning Guide.

OBJECTIVE: Develop a simple plan to specify who should talk, when, and via what medium.
☐ Check orders and SPINs, ROEs, pre-approved actions, etc. for applicable:
o Comm standards
o Check-in procedures
o Authentication
o Other comm procedures applicable to the mission
☐ Comm Plan Specifics:
o Weapon system / location operations
o Sortie Crew Shift / timing
o Chatrooms / channels (i.e., single chat room vs multiple chat rooms)
☐ Specific server names / URLs / IPs as needed
☐ Chat room assignments
☐ Room purposes (execution vs planning)
o Multi-level security considerations
o Triggers / contracts for transitioning between primary and backup comm